RISA LABS

AI Browser Assignment: Conceptualizing the Future of Browsing

Introducing

JARVIS

Shritik Jaiswal

IIT Kharagpur

Institute ID: 21BT3EP08

Phone: 91 9975343323

Mail: shritik@kgpian.iitkgp.ac.in

JARVIS: Reimagining the Future of Desktop Browser for an AI driven world

Hello, I am Jarvis.
Your ultimate guide.

Desktop Browsers were built for static use, while users are adapting to AI-assisted workflows. JARVIS aims to close this gap with an AI-native, task-aware browsing experience.

WHY Browsers?

- **275B / yr** — Browser-influenced value
- **87B / yr** — AI browser segment by 2030
- **30%** — AI browsers CAGR through 2030
- No dominant AI-native browser category

WHY Browser Market is Huge?

- **4.7 Billion** — Browser MAUs
- **32%** — Active Screen Time on Browser
- **2.7 Hr / Day** — Average User Time Spend on Browser
- **50+** — Webpage / Day visited by Average User

WHY the problem really matters?

NO Intelligence — *Traditional Browsers are just built for page rendering*

- Cannot detect user intent
- No grasp of complex content

Manual Work — *Everything needs human effort*

- Repetitive micro-actions daily
- No automated task flows

Why Tradtional Browsers are going off?

Zero Insights — *Pages shown, meaning extracted manually*

- No summaries or highlights
- No Multi-page comparison / Insight

Context Blackout — *Tabs exist, relationships don't*

- No tab intelligence or grouping
- No continuity across sessions

WHY Now?

- AI usage surging
- LLMs becoming reliable
- Cloud-first workflows

AI Browsing is NEW & Underserved

*Recent launch of **Atlas & Comet** aren't coincidences; they signal strategic shift toward AI-powered browsers.*

AI Browser Market Share

Traditional Browser	AI-based Browser
93%	7%

AI Browser Market Share Increased by 4% from 2024 to 2025

We surveyed UG /PG students early-career professionals to understand how they navigate complex online information and where today's browsers fail them.

Primary Research

- **178** Surveys
- **21** Interviews

Surveyed Profiles

- **72%** UnderGraduates
- **18%** Employees

Key Insights from Survey

- 58% struggle to understand complex content without instant AI explanations inside the page.
- 54% said AI summaries greatly reduce reading time but require switching tools constantly.
- 49% want the browser to remember context and continue tasks intelligently across sessions.
- 33% struggle to maintain context while juggling academic sources manually.

Word Cloud based on Primary Research

AI-Assisted Browsing Behavior Shift

AI doesn't replace browsing; it reshapes it. Users are shifting from manual, tab-heavy actions to assisted, efficient, task-driven behaviors. AI is becoming the first layer of interpretation.

Feature Usage	Decrease	Stay the Same	Increase	% of Respondent Net Intent	Explanation / Behaviour
Opening Multiple Tabs	62 %	35 %	3 %	↓ 59%	GPTs became go-to Platform
Reading Full Content	71%	26%	3 %	↓ 68 %	Summary-First Behaviour
Information Comparision	25 %	43 %	33%	↑ 8 %	Increased AI Comparisons
Performing Repetitive Searches	68 %	29 %	3%	↓ 65 %	Fewer Manual Searches
AI Summaries before reading	2 %	36 %	62 %	↑ 60 %	Summary-Driven Reading
Asking AI for Explanation	5%	38 %	53%	↑ 48 %	AI Clarification First
Using AI to Code	60%	35 %	5 %	↓ 55 %	Reduced Manual Debugging
Exploring Non-Essential Tools	74 %	22 %	4 %	↓ 70 %	Reduced Exploration

The % of Google searches that result in no clicks increased to 69% (May 2025),

The signals & data are clear: users changed, tasks changed, expectations changed. Now the browser must change too.

Who We're Building For? Understanding Our Core Users

Let's decide a focus group & understand their core needs

For who are we building first ?

UG Student   PG Student

User Segmentation

Age Group

- School Kids (0-14 Yr) - Low Digital Access
- College Students (14 - 25 Yr) - Highest AI Adoption (#1)
- Adults (26 - 60 Yr) - Huge Market Size but Fragmented (#2)
- Senior Citizens (60+) - Selective Browser Usage

Life Stage

- School - Books/resources focused (#2)
- Undergraduate - Knowledge gain (#1)
- Postgraduate - Depth focused (#1)

Behavioural Segmentation

Prioritized user segments — because their behaviours and painpoints represent the core problems shared across most user groups

- Readers & Researchers (PhD student, Medical student)
- Multi Tab Navigators (MBA student solving case studies
- WorkFlow Users (Product Intern using Sheets, Docs)

Low Priority / less browsing usage

- Tool Centric Users
- Transaction User
- Entertainment / Media Consumers
- Miscellanous Users

Shivang, B.Tech (Biotechnology), 21 Yrs - A Multitasking, fast-switching learner

GOAL: To understand complex science topics quickly and complete assignments with clear conceptual clarity.

Habits

- Opens 20–30 tabs during research sessions
- Switch b/w GPT, Google, NPTEL, Sci-Hub, Quora
- Watches YouTube lectures, reads PDFs + Notes
- Saves screenshots / random links to revisit

Needs

- AI explainers directly on the page / pdf
- Smart tab organization for assignments
- Quick summaries of PDFs, research articles
- Personal AI companion for explanations

Painpoints

- Struggles to understand dense scientific content
- Constant lookup of terminology breaks study flow
- PDFs are overwhelming and take too long to study

"I jump between so many pages just to understand one concept… it's exhausting. I wish the browser could explain things to me without breaking my rhythm."

Srishti, PHD (Nano Technology), 31 - A detail-oriented, evidence-driven researcher

GOAL: To synthesize insights from multiple research papers and build clear arguments for experiments and publications.

Habits

- Work on papers, citations, datasets, SEM/TEM
- Reads large PDFs, materials, lab manuals, etc
- Uses AI Summary but cross-verifies manually
- Maintains insight notes on Notion, Docs, Sheets

Needs

- Multi-resource, Reliable summary
- Seamless linking b/w paper, citations, notes
- In Depth explainer of jargons ,equations, term
- Organized research pipelines

Painpoints

- Scientific PDFs are too dense & time-consuming
- No tool helps connect insights across tabs or papers
- Difficult to manage long-term research threads and keep continuity

"My research is scattered across dozens of papers. If the browser understood my research workflow, half my burden would disappear."

User Journey of a College Student (UG / PG)

L-0	Pre-Study				During Study					Post-Study	
L-1	Knowledge Requirement	Search	Evaluate	Select Sources	Read	Verify	Compare	Note / Learn	Write / Finalize	Organize	Repeat Learning Continuity
L-2	Read Offline Resources (Book / PDFs) → Realize Knowledge Need	Open Browser → Initiate Google Search	View 10-20 Links and Media Options	Open Top Links to get information	Read & Scroll Webpages	Switch between 3–10 tabs for more knowledge, comparision and verfication		Note Insight on Docs, Notes, etc	Learn write assignments, reports, research briefs	Organize Data, Insights, Finding, references, Notes	Revisit earlier links to refresh memory → Search for missing pieces → loop
Pain/Need	Need explanation before searching	Query repetition	Can't judge without opening leading to Tab Overload	Difficult to decide, start with 1st Link	Consumes Time & Effort, Difficult to understand	Tab switching fatigue, kills focus	Slow and Error Prone	Scattered / Broad / Vague notes	Struggle to remember, lose context	Scattered insights, tedious task; citations are messy.	Lost continuity, reopens tabs, repeats searches
Solution	*PDF/Notes Default Drive* *Reading Companion*	*AI Query Assist*	*Summary on Search for Each Link*	*Chat driven suggestions* *Progress Organizer*	*Deep Explainer Auto-Highlight* *Reading Companion*	*Split-Screen Suggestion, Passive Tab Insights* *Memory-Strip Mode*	*Task Suggestion by user intent* *(Task = Comparision in this case)*	*AI Notes Panel,* *Raw Data to Insight Generator*	*Writer Assistant*	*Smart Organizer Suggestion + Executor Auto Citation*	*Session Memory Visualization*

How JARVIS Reinvents Browsing for Student Workflows

Let me streamline your entire study workflow

Introducing JARVIS Browser — an AI-native workspace built to rethink how students research, learn, and create.

1 **JARVIS Homepage - Overview**

Acts as a unified workspace where user begin / resume tasks with clarity. It centralizes search, tasks, folders, quick-access tools, and AI assistance to minimize friction.

UX RATIONALE: Designed Around Natural User Behaviour

- Search is placed at the center because users naturally focus on the center of the screen first.
- A sidebar replaces traditional top-tab navigation to maximize screen space, improve organization, and support drag-and-drop folder management.
- Quick tools and AI access remain within thumb/eye reach, creating a behavior-aligned browsing workflow.

Key Experience Improvements

- Instant access to recent work, research folders, reminders
- Personalized quick access tabs reduce repetitive navigation
- AI available immediately from homepage
- Smooth transition into study mode or project workflows

High Level Workflow

Open JARVIS → View Quick Tabs / Notes / Central Workspace → Access Saved Tabs / Ask JARVIS / Initiate Search

Access Saved Tabs → Customized Saved Tabs / Sidebar

Initiate Search → Land JARVIS / Result Page

HOMEPAGE

Search TAB at Center
Drive UX Principle

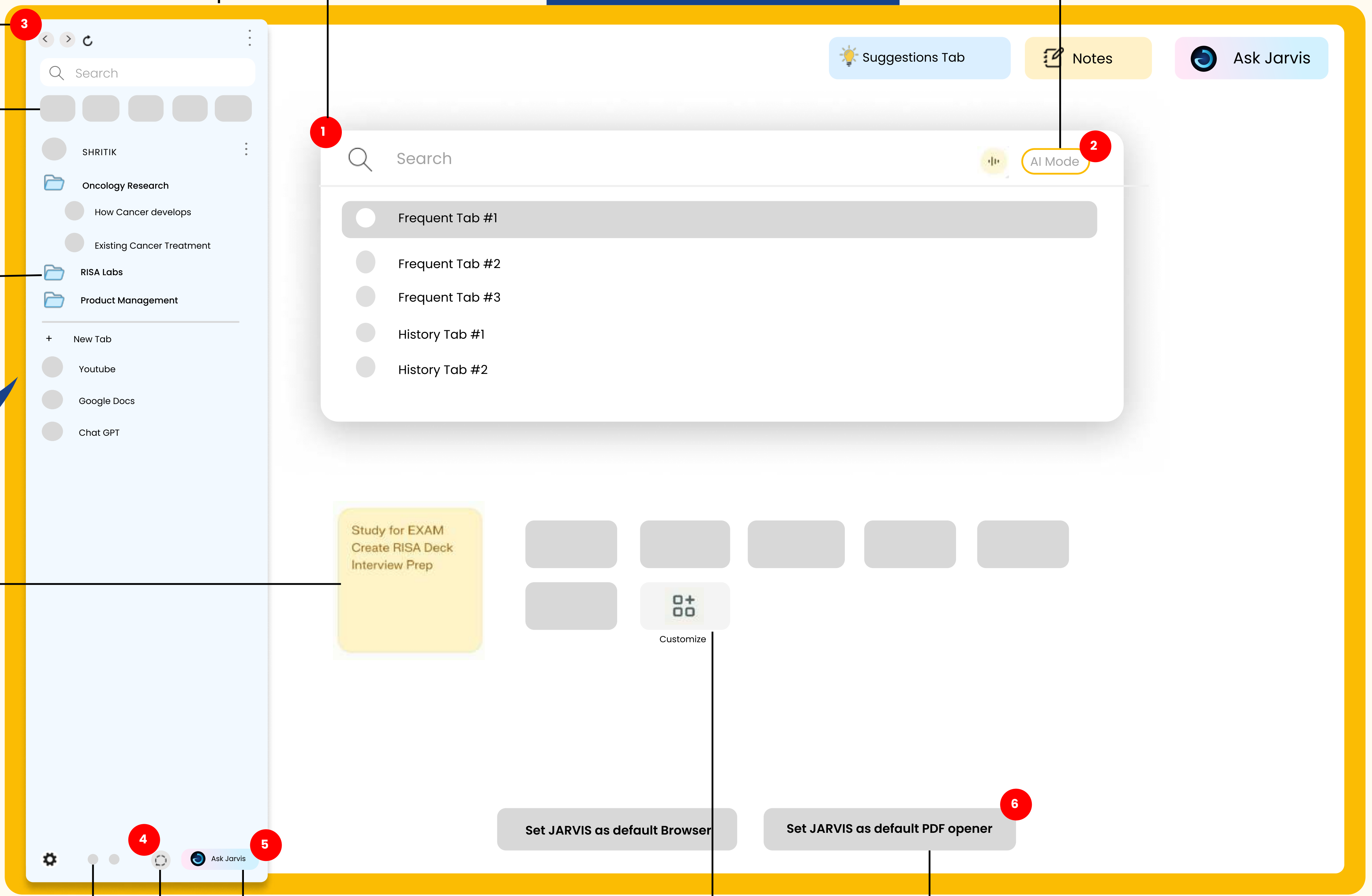
AI Search Enable / Disable
Gives control to Browse simply

Side Panel
Eliminates Cognitive Load

Bookmarks

Folders

Notes & Task Reminder



JARVIS Quick Access - *Always at Front*

Default Browser Drivers - *Increase Adoption*

Quick Access Tab Customizer

Quick Access
Profile Switch UX
Drives Profile Seperation

Temporary Chat Mode - *No Memory of Conversations*

Research didn't evolve with students—so JARVIS evolves the browser instead.

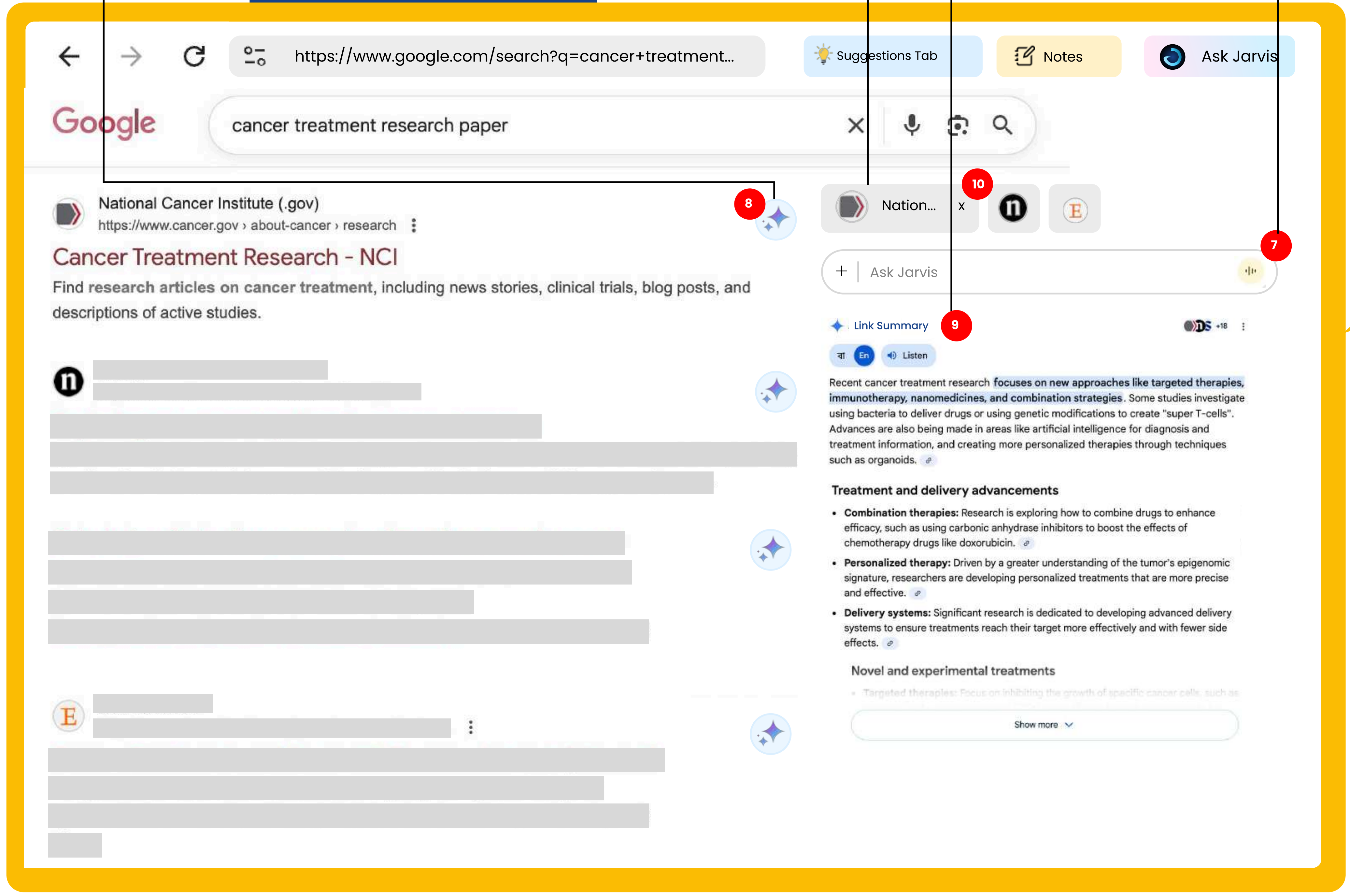
RESULT PAGE

Summarization button - Tells "What's Inside Link?"

Shows "What's Inside Link?" - Summary

Summary Organizer
-via Tabs

Interactive Browsing



2 **Evaluate Sources Without Opening Tabs - on Result Page**

AI-enhanced search that summarizes each link from the results page, organizes clicked links into tabs, and enables follow-up Q&A; helping users evaluate sources quickly without opening unnecessary pages.

High Level Workflow

Link Summary Button → Page Summary Generation → JARVIS Follow-Up Chat

Page Summary Generation → Other Link Summary → Link Wise Summary Tab Organization

Other Link Summary → JARVIS Follow-Up Chat

Paipoints Addressed

- Hard to judge link relevance quickly
- Wasted time opening irrelevant pages
- Frequent context switching breaks flow

Key Experience Improvements

- Preview links instantly via AI
- Faster selection of high-value sources
- Ask follow-ups w/o opening pages
- Quicker, more confident decision-making

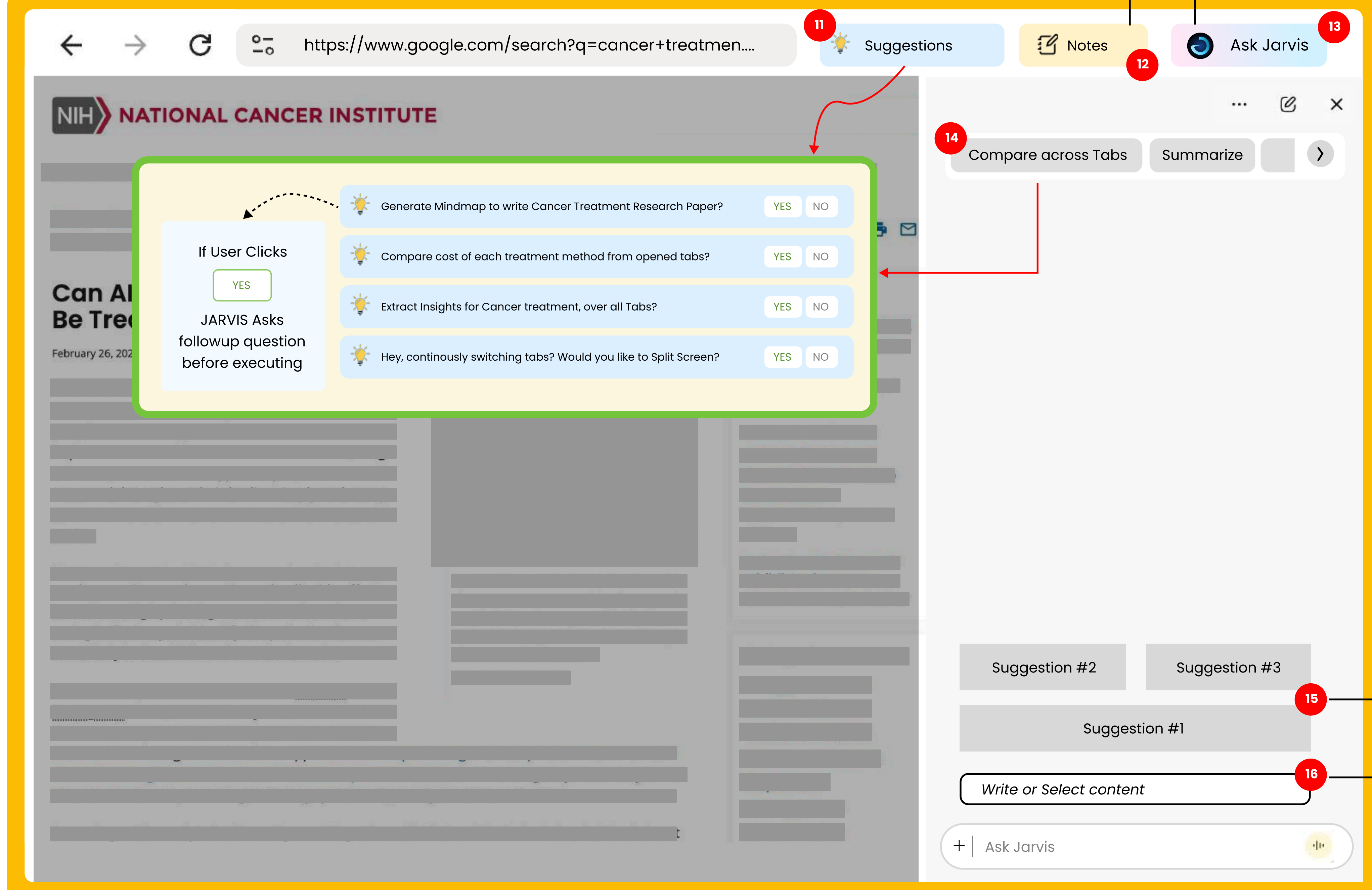
Transforming Research with Intelligence Companion

Let me guide you during your research

JARVIS SIDE PANEL AND HEADER FLOATS

All time accessible Notes

JARVIS AI Quick Access - *Always float on top right*



Prompt Suggestions - *Drives Usability*

Informs JARVIS takes inputs through content selection too.

3

Intelligent Assistant: Turning Complex Research Into a Guided Flow

JARVIS intelligently detects user intent, suggests relevant actions, offers AI-organised notes, and provides an always-visible Ask-JARVIS panel. Together, these features guide users through complex tasks, reduce manual effort, and keep research flowing smoothly.

High Level Workflow

WebPage / PDF Load → Intent Detection → Shows Smart Suggestions

Shows Smart Suggestions → Follow-Up Questions → Task Execution → Note Insights

Note Insights → Research Continuity → JARVIS interaction (Anytime) → Follow-Up Questions

Paipoints Addressed

- Manual effort to extract & organise insights
- Difficult decision making
- No central place for notes and AI companion

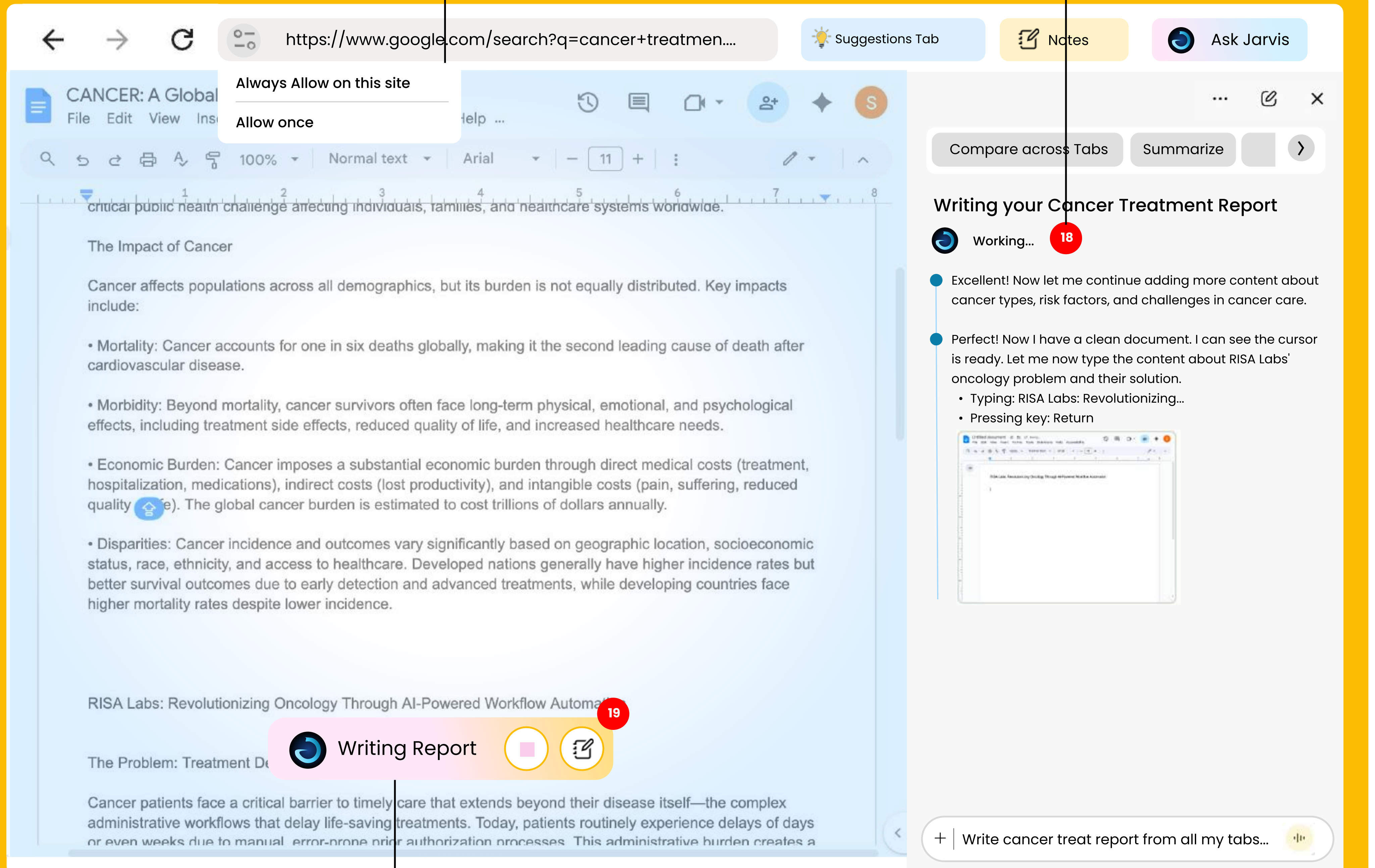
Key Experience Improvements

- Smart user intent based suggestion
- AI-organised notes for quick recall
- Always-available Ask Jarvis panel
- Smooth transition between actions

Workflow Automation

Permission Access

Progress Summarizer



Automated Task Controller - *Allows stoping / Editing the task*

4

Turn Complex Tasks Into One-Click Actions

JARVIS automates multi-step research and writing tasks by taking controlled actions on the screen—clicking, extracting, typing, and compiling information from all open tabs to produce a ready-to-edit document with user permission and real-time progress feedback.

High Level Workflow

Trigger Action via Prompt / Suggestions → Tab Access Request → Perform Actions

Perform Actions → Show Live Steps → User Pause / Edit Action anytime → Output

Output → Enhance Results / Reprompt → Perform Actions

Use Case Examples

- Generate Multi-Source Reports & Assignments
- Fill Forms / Data Automatically
- Execute Calculations & Formulas in Google Sheets
- Write Mails, Reports, Assignments inside Docs
- Highlight PDF while reading
- Perform Benchmarking Product and note comparisions

Priorities, Differentiators & Partnerships

Let's Align Strategy
Roadmap for our users

RICE Feature / Solution Prioritization

Feature	Reach	Impact	Confidence	Effort	Score	Rank
Temporary Chat Mode	8	7	9	4	126	#4
Default Browser Driver	7	9	5	3	94	#8
Side Panel Workspace	10	9	8	6	120	#6
Result Link Summary	10	10	9	6	150	#2
Summary Tab Organizer	7	6	8	6	56	#10
JARVIS Interactive Chat	10	10	8	6	134	#3
Intent-Aware Suggestion	10	10	7	8	88	#5
AI Notes Panel	8	7	8	7	64	#9
Workflow Automation	10	9	7	5	126	#5
Permission Control	10	10	10	3	333	#1

Layered Value Stack of JARVIS Browser

JARVIS Core Value Features

- ASK Jarvis
- Permission Control
- Side Panel
- Result Link Summary

Support Systems

- Smart Suggestions
- Temporary Chat Mode
- Tab Organizer
- Result Link Summary

Delighters

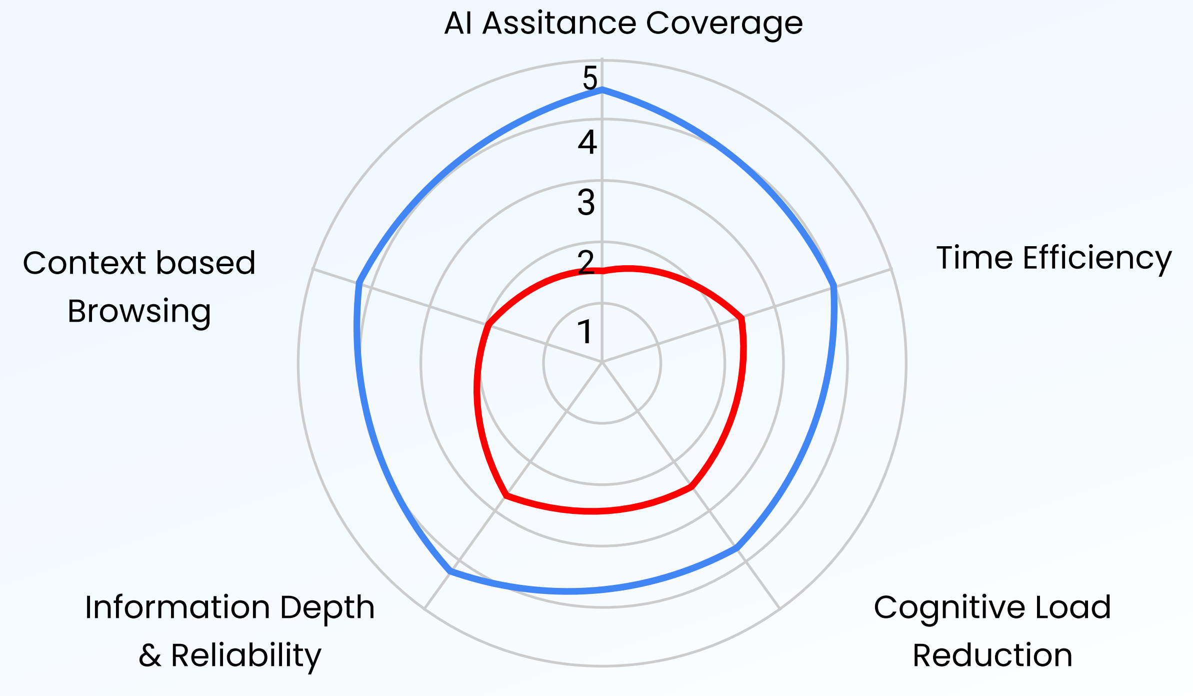
- Workflow Automation
- AI Notes Panel

How JARVIS Differentiates from existing Browser

Clear Functional Differentiators

- One-Click Source Understanding
- Context Intelligence & Smart suggestions
- Cross-page passive insights
- Self-Organizing Research Space
- Task Level Automation
- Study Companion - Deep Explainer

User Outcome Differentiators



User Emotion Differentiators

"The suggestions feel uncanny — JARVIS knows I'm switching tabs too much and offers a split screen before I even realize the problem."

"The tab organizer and notes AI keep everything clean — I don't feel that mental chaos that normally hits after 30 minutes of research."

"When I'm stuck, JARVIS explains things like a patient tutor. It feels like someone's actually helping me, not just a browser."

Strategic Partnerships & Integration to Accelerate JARVIS Adoption

By solving the right problems first, JARVIS becomes an indispensable study partner

1. Academic PDFs & Research Content Platforms

(arXiv, PubMed, Semantic Scholar, Sci-Hub mirrors)

Why this Matters

65% *Student's study time spent on PDFs, papers, slides*

42% *Fnd academic PDFs hard to understand alone*

Unique Values Created:

- *AI Study Mode: Summaries, explainers, highlights from any academic PDF.*
- *Instant Q&A: Ask doubts directly on the PDF without switching tools.*

How JARVIS Drives Adoption

- *Campus rollouts: Partner with IITs, NITs, universities — "free AI PDF mode for students."*
- *PDF-first onboarding: Every PDF opens in JARVIS Study Mode → instant "aha" moment.*

2. EdTech & Learning Tools Integration

(Coursera, Udemy, Khan Academy, Unacademy, YouTube Lectures)

Why this Matters

4 hrs/day *Avg time by students on these sites during exams*

72% *Students rely on online lectures for learning course*

Unique Values Created:

- *Lecture-to-Notes Mode: Auto-generates clean, structured notes while watching any lecture.*
- *Instant Concept Explainer: Breaks down formulas, code, or theory inside the video; no tool switching.*

How JARVIS Drives Adoption

- *YouTube Notes Panel: Right-panel prompt → "Generate notes for this lecture?" → high-visibility entry point.*
- *EdTech Creator Collabs: Influencers & educators demonstrate JARVIS's AI benefits on popular courses.*

3. Productivity & Note Ecosystem Integration

(Notion, Google Docs, Sheets, Obsidian, Canva)

Why this Matters

85% *Students complete reports assignments on these sites*

56% *Employees / Productive Users use these platforms*

Unique Values Created:

- *AI Split-Screen Workspace: Research, summaries, and writing in one synchronized interface.*
- *Auto-Sync Insights: Summaries, citations, and highlights flow directly into Docs/Notion.*

How JARVIS Drives Adoption

- *One-Click Export: "Send summary to Notion/Docs" → direct workflow integration.*
- *Student Templates: AI-ready templates for lab reports, literature reviews, and case study drafts.*

Building a Student-Focused Revenue Flywheel

You study better;
I grow stronger with you!

Go-To-Market Strategy

Target Areas		Campus clubs & Department Groups	Undergraduate Students	Post Graduate Students
AWARENESS	Goal	Create high visibility within academic communities.	Reach students during assignments and study hours.	Target heavy research users facing daily PDF pain.
	Strategy	Run AI workshops using real research tasks; Collaborate with faculty and club leaders.	Instant AI summaries for difficult topics. Use creators to show live research workflows.	AI explainers for dense research papers. Show multi-tab insights for reviews.
	Channels	Workshops • Faculty • Club Leads	YouTube • Instagram • Search	Research Labs • Dept Groups • LinkedIn
	Why it works	Faculty demos create quick credibility; Student clubs quickly drive tool adoption.	UGs seek fast clarity when confused. Creators strongly shape student tool choices.	PGs struggle with complex multi-source reading. AI cuts research time significantly.
ACQUISITION	Goal	Convert active members through hands-on usage.	Enable frictionless try-outs during study sessions.	Acquire research-heavy users with high-value tools.
	Strategy	Offer beta access for research workflows; Run live demos with department PDFs.	Use Chrome extension as main entry point; Show AI summary button on search results.	Lead with AI PDF explain mode; Enable multi-tab summaries for deep reviews.
	Channels	Beta Access • Live Demos • Labs	Extension • Search Banner • Docs Sidebar	PDF Mode • Tab Summary • Faculty Mailers
	Why it works	Hands-on trials build strong conviction; Department tasks align well with JARVIS.	UGs prefer zero-effort onboarding. Instant summaries give immediate value.	Directly targets PG research pain; Boosts multi-source comprehension speed.
ACTIVATION	Goal	Ensure first WOW moment in live sessions.	Deliver instant clarity within first 30 seconds.	Activate PGs through deep research assistance.
	Strategy	Demonstrate split-screen research workflows instantly; Show tab grouping and quick summaries.	Explain PDF sections with one-click AI; Organize tabs automatically during research.	Provide insights across all selected papers; Enable JARVIS to draft structured notes.
	Channels	Workshops • Lab Hours • Club Meets	PDF Prompt • Summaries • Ask Jarvis	AI Notes • Insights Mode • Research View
	Why it works	Immediate value visible within seconds; Workflows align with department tasks perfectly.	UGs are overwhelmed by dense content; Automation reduces confusion and speeds learning.	PGs need strong multi-source synthesis; Structured outputs reduce manual research workload.

REVENUE MODEL

Freemium Core (Primary Model — 95% Users)

What Students Get for Free

- AI Summaries of webpages
- Link Preview Summaries
- Note Panel (basic)
- Ask-Jarvis (limited queries/day)
- Basic Split Screen
- PDF Reader (non-AI)
- Limited Smart Suggestions

Why This Works

- Students try without friction
- Drives high adoption leading to habit loops

JARVIS Pro — Student Subscription (₹99 / month)

Premium Features Included

- Unlimited AI depth explanations
- Unlimited Automated Workflows (form, report, Sheet, transactions, etc)
- Advanced Smart Suggestions
- Longer term context memory for research
- AI Workspace Templates (lab reports, literature reviews, case studies, PPT / Deck components)

Why This Works

- Affordable Price for Students
- High Value for Power Users
- Better Quality Results

Pay-Per-Use (Micro-transactions ₹10–30)

Examples

- Generate Lab Report Draft from All Tabs
- Auto-Solve Sheet Formulas / Data Cleaning
- Compare 10 Research Papers
- Extract Citations Automatically

Why This Works

- Students avoid subscription fear
- Only pay when needed
- High-Volume micro-utilities

Ethical First-Party Data Intelligence

What we Monetize

- Academic intent & topic trends
- Study behavior & content / browsing patterns
- Skill interests & course demand
- High-intent learning segments

Why This Works

- Browsers hold rich, high-intent user behaviour
- Student focused Product access target audience

Let's elevate student learning together
with AI that works beside you always